Project- Cisco Packet Tracer

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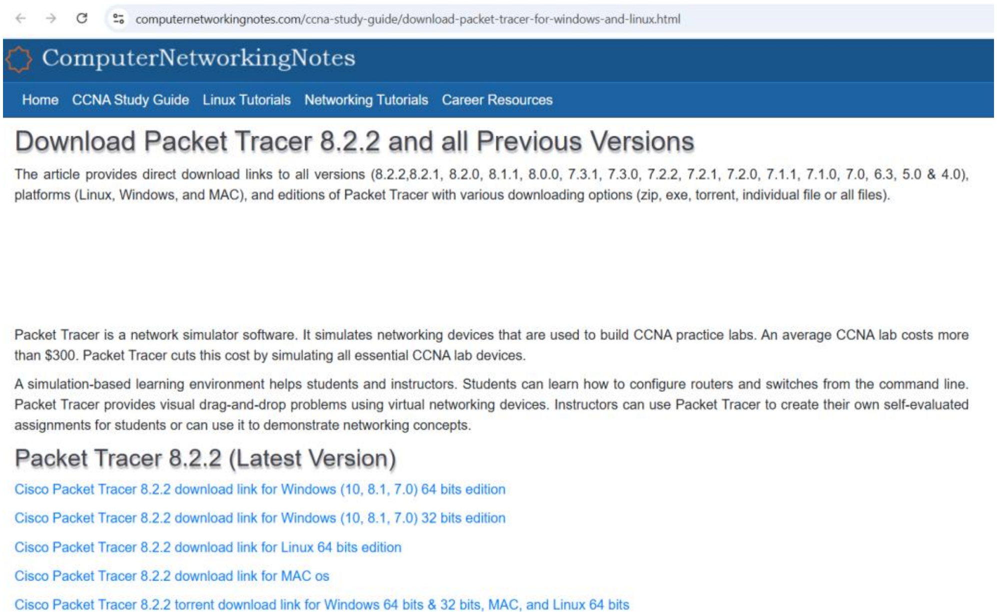
Cisco Packet Tracer

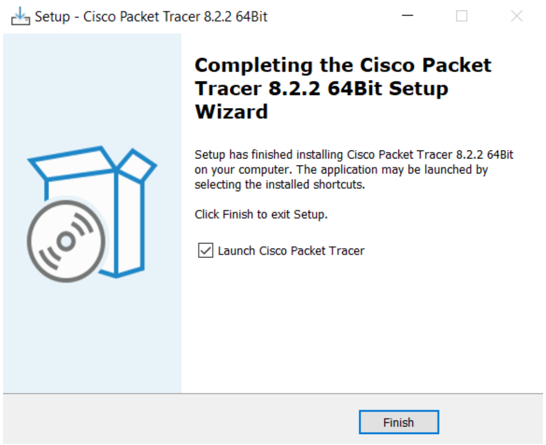
Introduction

Cisco Packet Tracer is a simulation tool created by Cisco that lets you build and test computer networks. It helps you design different network setups and see how they work without needing real hardware.

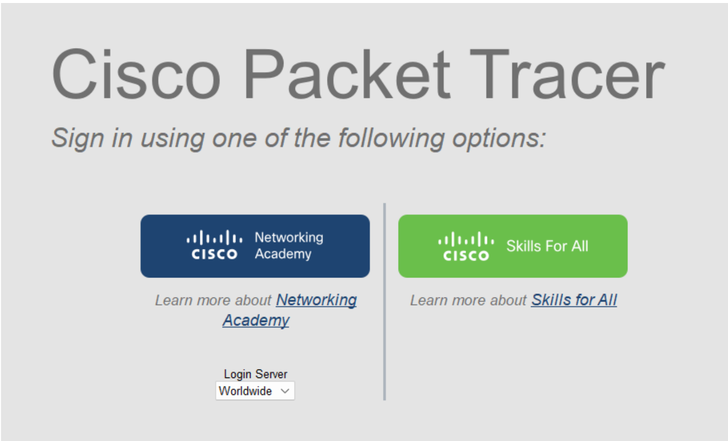
Step 1: Go to the website: https://www.computernetworkingnotes.com/ccna-study-guide/download-packet-tracer-for-windows-and-linux.html and download the Cisco Packet Tracer that is appropriate for your OS.

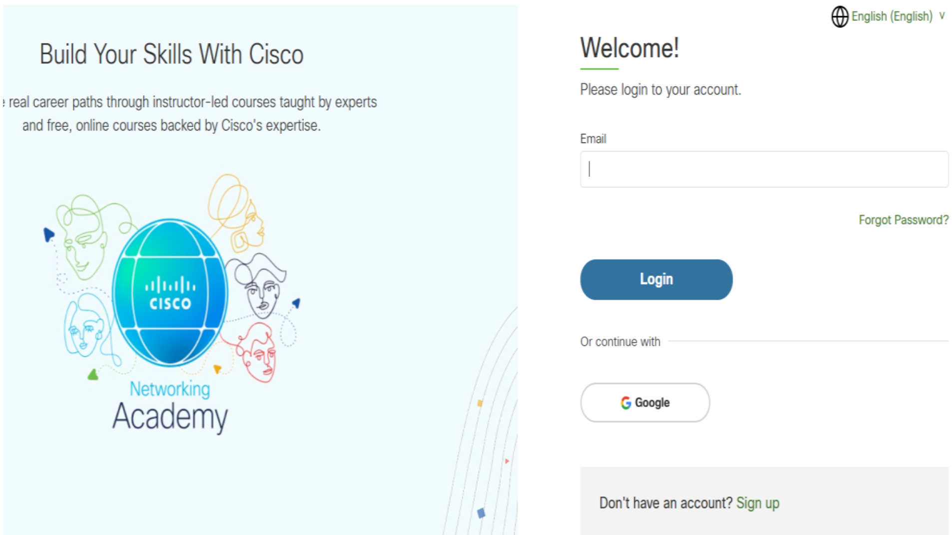


Step 2: Run the installer file and follow the prompts to complete installation. Once installed, locate the application on your computer and click finish to open it.



Step 3: You will be prompted to log in using your Cisco Networking Academy credentials or you will need to create a new Cisco Networking Academy account. Complete all of the registration information and confirm to the terms and conditions. Click Accept and Continue, answer what your practical experience is in IT, networking and cybersecurity and click create account.

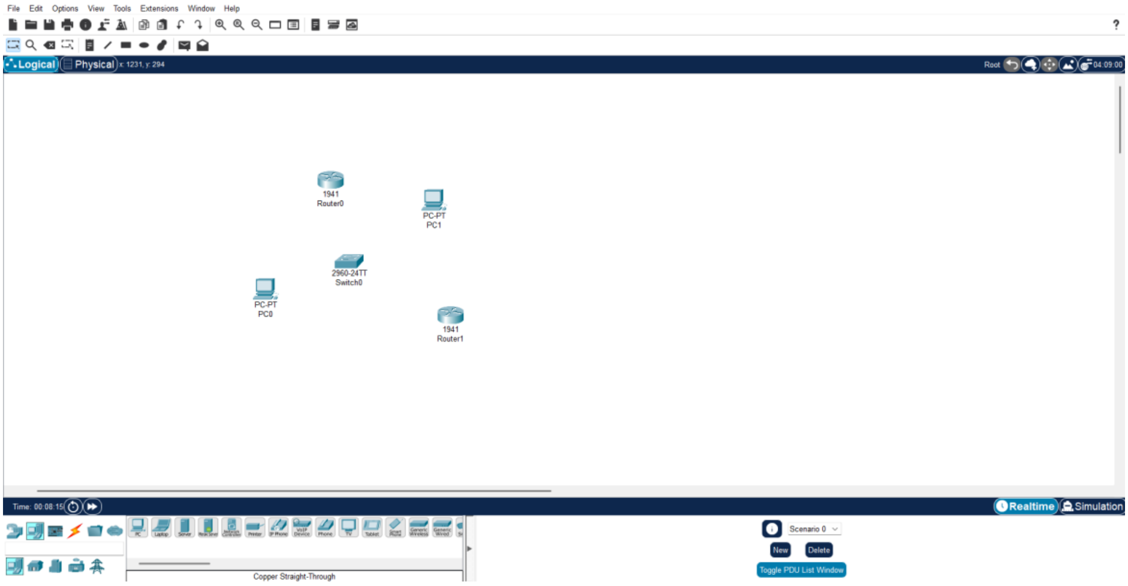




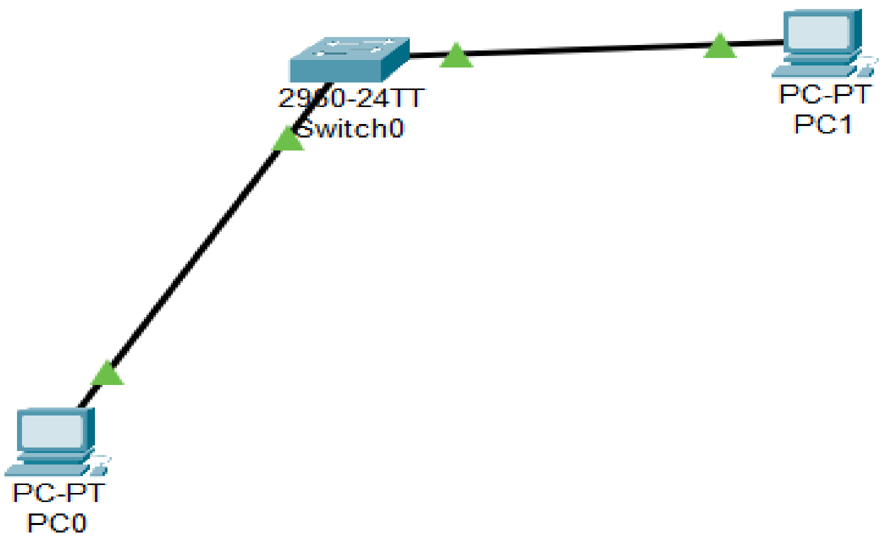
Set up a Basic Network

I will build a basic wireless network.

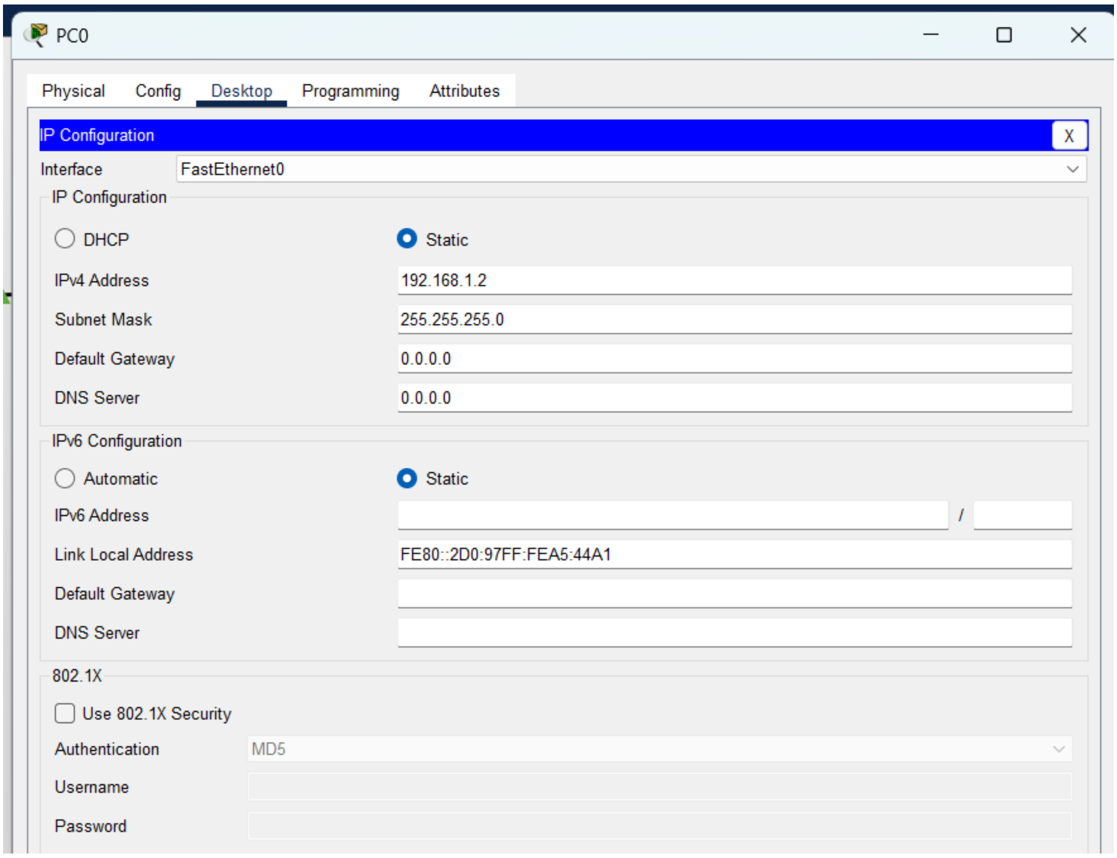
Step 1: In the workspace, drag routers, switches, and PCs from the device selection menu at the bottom left of the page.



Step 2: Connect the devices by using the connection tool (represented by a cable icon) to connect the devices together with appropriate cables (e.g., copper straight-through for PC to switch) and connecting a switch and two PCs together.



Step 3: Configure device IPs by clicking each PC or router to access the settings. Assign IP addressees for the PCs, navigate to "Desktop" tab and click "IP Configurations" to assign and IP address and subnet mask. Ensure all devices are on the same network by configuring IPs with the same subnet (e.g., 192.168.1.2 for PC1, 192.168.1.3 for PCs, etc.).



Step 4: Test the network. Use the ping tool by right clicking on a PC and click on the "Command Prompt" on any P and use the "*ping*" command to test connectivity (e.g, *ping 192.168.1.20* to ping another PC).

***If the ping is successful, the network is configured correctly

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Pinging 192.168.1.20 with 32 bytes of data:

Reply from 192.168.1.20: bytes=32 time=34ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 34ms, Average = 8ms
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Step 5: Save your network by clicking on the "File" menu and select "Save As". This ensures you can reload the network setup later for further testing.

Conclusion

This project taught me how to install and troubleshoot using Cisco Packet Tracer. I understand the interface and tools for creating networks. I can assign IP addresses and configure devices for communication. I can run ping tests to verify network communication between devices. Saving and exporting network topologies in *.pkt* format to share with others or to continue working in different environments.